```
module vending_machine(clk, rst, x, out, change);
 input clk, rst;
 input [1:0] x;
 output reg out;
 output reg [1:0] change;

 reg [1:0] p_state, n_state;

 parameter S0 = 2'b00;
 parameter S1 = 2'b01;
 parameter S2 = 2'b10;

 always @(p_state or x)
 begin
        n_state = S0;
                change= 2'b00;
         case(p_state)
         S0: if(x == 2'b01) begin                          //Received 5 Rs
                   n_state = S1;
                   change = 2'b00;
                   out = 1'b0;
                  end
                  else if(x == 2'b10) begin                //Received 10 Rs
                   n_state = S2;
                   change = 2'b00;
                   out = 1'b0;
                  end
                  else begin
                   n_state = S0;
                   change = 2'b00;
                   out = 1'b0;
```

```
end
S1: if(x == 2'b01) begin                    //Received 10(5+ 5) Rs
  n_state = S2;
  change = 2'b00;
  out = 1'b0;
end
else if(x == 2'b10) begin                   //Received 15(5+10) Rs
  n_state = S0;
  out = 1'b1;
  change = 2'b00;
end
else begin
  n_state = S0;
  change = 2'b01;
  out = 1'b0;
end
S2: if(x == 2'b01) begin                        //Received 15(5+ 5 + 5) Rs
  n_state = S0;
  out = 1'b1;
change = 2'b00;
end
else if(x == 2'b10) begin                       //Received 20(10+ 10) Rs
  n_state = S0;
  out = 1'b1;
  change = 2'b01;
end
else begin
  n_state = S0;
  change = 2'b10;
  out = 1'b0;
end
```

```
                endcase
        end
        always @(posedge clk or posedge rst)
         if(rst)
           p_state <= S0;
    else
           p_state <= n_state;

endmodule
```